

Table 10: **Scaling experiment results (length-controlled win-rates) on AlpacaEval 2.0.** We put different amounts of random examples drawn from SkillMix to the context as demonstrations. We conduct experiments on two base models: Mistral-7B-v0.2 and Meta-Llama-3.1-8B. The results are computed over 3 different runs.

	AlpacaEval 2.0 LC Win-rate (%)	
	Mistral-7B-v0.2	Meta-Llama-3.1-8B
0 example	2.92 ± 0.0	2.99 ± 0.0
1 example	11.98 ± 0.7	8.66 ± 2.7
3 examples	12.50 ± 1.3	11.56 ± 0.6
10 examples	14.86 ± 0.4	15.11 ± 1.9
20 examples	13.20 ± 0.7	15.18 ± 0.8
30 examples	12.45 ± 0.1	13.90 ± 0.2
40 examples	12.51 ± 1.3	14.95 ± 0.7
50 examples	13.80 ± 0.7	15.49 ± 0.8

E ADDITIONAL EXPERIMENTS

E.1 SCALING EXPERIMENTS ON ALPACAEVAL 2.0

This section provides additional results of scaling experiments conducted on the AlpacaEval Suite, which contains 805 single-turn examples, an order of magnitude more than MT-Bench. Based on the SkillMix dataset, we construct the set of in-context demonstrations by sampling different amounts of random examples, ranging from 0 to 50 examples. The results shown in Table 10 confirm our findings, reported in Sec. 3, that scaling the number of ICL examples is not sufficient to consistently improve the instruction following performance.

E.2 GREEDY SEARCH ON MISTRAL-7B-v0.2

In this section, we show the distribution of the resulting scores (with GPT-4-Turbo as the judge) for each step of the greedy search on SkillMix-4k dataset when using Mistral-7B-v0.2 as the base model. Similar to the finding on Llama-3.1-8B model (see Sec. 3.2), the majority of demonstrations increase the alignment performance, as measured by 1st-turn MT-Bench score, when added as the fourth example on top of URIAL, and few progress is made with more demonstrations. We show more details of greedy search results on Mistral-7B-v0.2 in Fig. 12.

E.3 MORE EXPERIMENTS ON DECODING SCHEMES

We show more results of decoding schemes under different settings in this section, as supplementary to the main results in Sec. 2.2. Specifically, in Fig. 13, we turn off the repetition penalty (i.e., repetition penalty = 1.0) and show the 1st-turn MT-Bench scores on Mistral-7B-v0.2 when varying the values of temperature and top-p. Moreover, we repeat the same experiment procedure but on a different base model, Llama-3-8B, and show heatmaps in Fig. 14.

E.4 A BREAKDOWN EXAMINATION OF URIAL-ALIGNED MODELS

The test set of MT-Bench (Zheng et al., 2023) is comprised of high-quality questions that belong to 8 common categories: coding, math, reasoning, extraction, roleplay, writing, humanities/social science, and STEM. In Table 11, we present the per-category performance on MT-Bench for each model in order to have a more detailed comparison between the URIAL-aligned models and the corresponding instruct models. The results suggest fine-tuning is almost always better than URIAL (ICL) and only underperforms in a minority of cases on particular base models, such as Mistral-7B-v0.1 and Llama-2-70B.

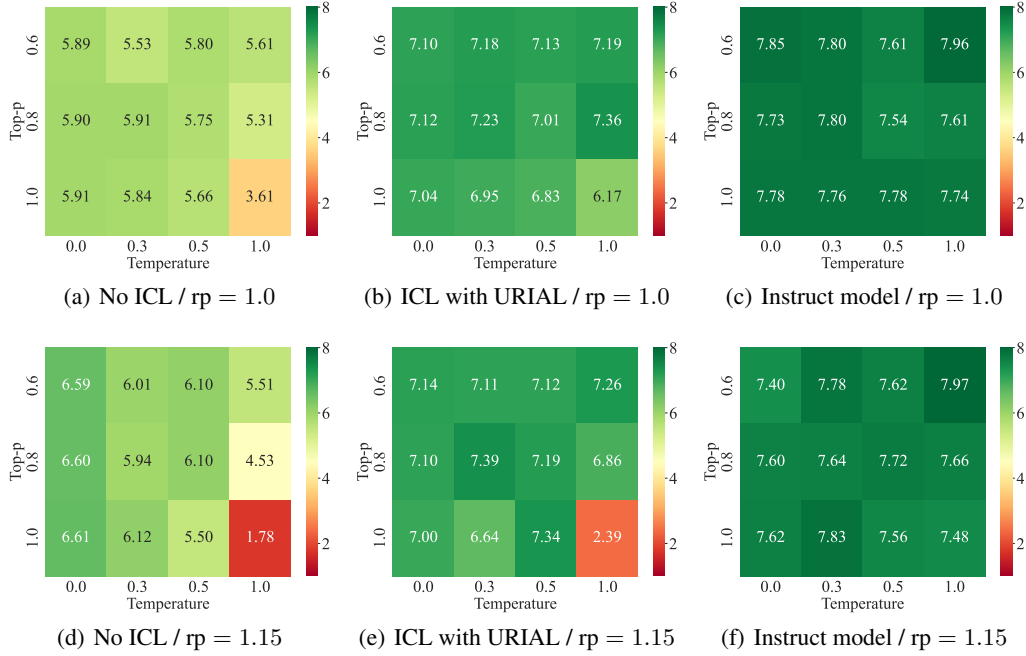


Figure 13: **The 1st-turn MT-Bench scores of model generations with and without ICL across different decoding schemes.** We mainly consider two hyper-parameters: temperature and top-p. The heatmaps show the answering quality of the base model with and without URIAL in the context is sensitive to the setups of decoding schemes. Surprisingly, with proper decoding parameters, the base model alone is already capable of following instructions.

Table 11: **A breakdown examination of URIAL-aligned models and corresponding instruct models across different categories on MT-Bench.** * denotes the results taken from the URIAL GitHub repository.

Model	coding	math	reasoning	extraction	humanities	roleplay	stem	writing	Average
Llama-2-7B + URIAL *	1.65	1.60	3.45	3.40	8.08	7.48	6.80	6.20	4.83
Llama-2-7B-Instruct	2.95	2.40	5.20	6.33	9.58	7.83	8.88	9.05	6.53
Llama-2-70B + URIAL *	4.15	3.60	6.10	7.70	9.75	7.33	8.75	9.50	7.11
Llama-2-70B-Instruct	3.75	4.10	5.95	7.40	9.85	7.90	9.13	9.50	7.20
Llama-3-8B + URIAL *	4.15	2.60	3.50	5.25	8.90	7.30	8.15	6.13	5.75
Llama-3-8B-Instruct	5.95	5.05	6.15	9.16	9.90	9.05	8.95	8.70	7.86
Llama-3-70B + URIAL *	4.35	3.80	4.95	6.20	8.00	7.25	8.55	8.10	6.40
Llama-3-70B-Instruct	7.85	7.35	6.25	9.75	10.00	9.30	9.60	9.80	8.74
Llama-3.1-8B + URIAL *	4.35	3.25	3.95	5.95	9.00	6.95	8.00	7.60	6.13
Llama-3.1-8B-Instruct	6.40	6.50	5.70	8.78	9.80	9.00	8.60	9.20	8.00
Mistral-7B-v0.1 + URIAL *	4.60	3.40	4.90	7.75	9.08	7.65	8.28	7.75	6.67
Mistral-7B-Instruct-v0.1	4.35	3.95	6.30	6.75	9.45	7.45	7.70	8.85	6.85
Mistral-7B-v0.2 + URIAL *	3.80	3.35	4.50	7.45	8.95	6.70	7.43	7.98	6.27
Mistral-7B-Instruct-v0.2	5.45	3.40	6.50	8.50	9.90	8.65	9.30	9.40	7.64
Mixtral-8x22B-v0.1-4bit + URIAL	5.25	5.85	6.00	9.20	9.80	8.65	8.30	8.60	7.71
Mixtral-8x22B-Instruct-v0.1-4bit	7.10	7.03	7.00	9.10	9.65	8.90	9.65	9.70	8.52
GPT-4-Base + URIAL	5.55	5.98	6.90	8.20	5.93	6.90	6.45	6.10	6.50
GPT-4 (March 2023) *	8.55	6.80	9.00	9.38	9.95	8.90	9.70	9.65	8.99

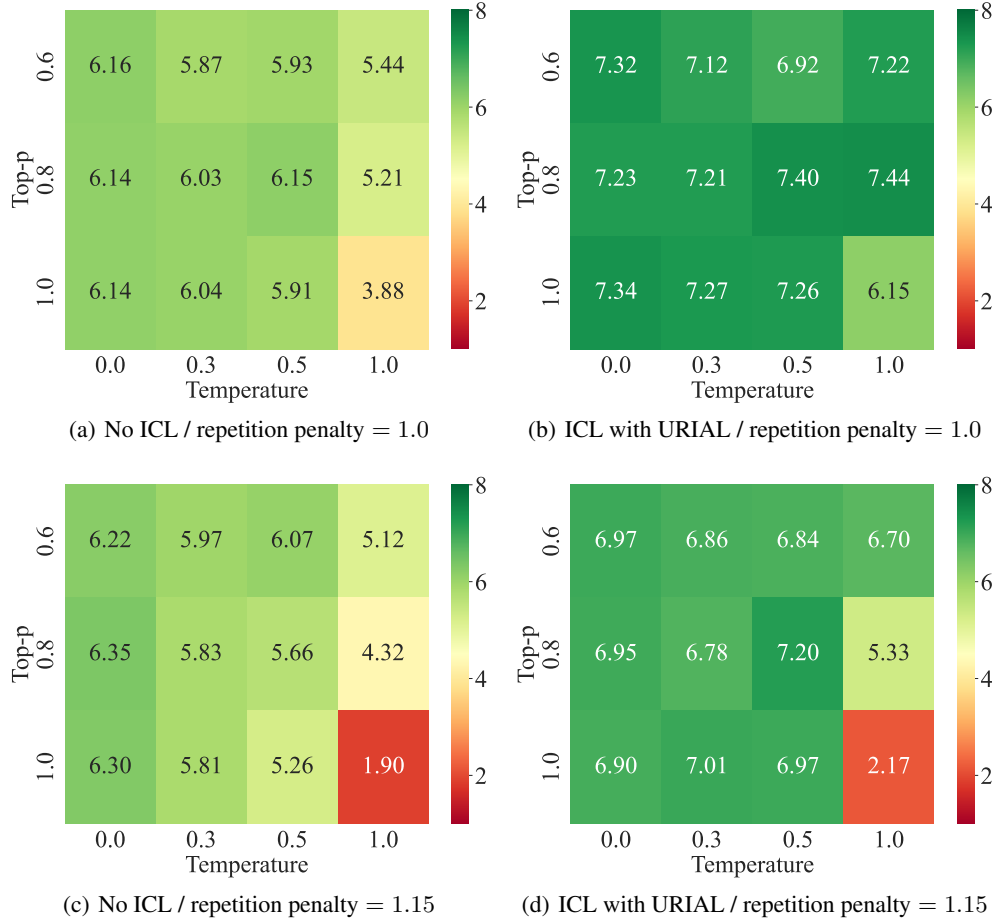


Figure 14: **The 1st-turn MT-Bench scores of Llama-3.1-8B generations with and without URIAL in the context across different decoding schemes.** We mainly consider two hyper-parameters: temperature and top-p.